

Neurobehavioral effects of aspartame consumption.



Despite its widespread use, the artificial sweetener aspartame remains one of the most controversial food additives, due to mixed evidence on its neurobehavioral effects. Healthy adults who consumed a study-prepared high-aspartame diet (25 mg/kg body weight/day) for 8 days and a low-aspartame diet (10 mg/kg body weight/day) for 8 days, with a 2-week washout between the diets, were examined for within-subject differences in cognition, depression, mood, and headache. Measures included weight of foods consumed containing aspartame, mood and depression scales, and cognitive tests for working memory and spatial orientation. When consuming high-aspartame diets, participants had more irritable mood, exhibited more depression, and performed worse on spatial orientation tests. Aspartame consumption did not influence working memory. Given that the higher intake level tested here was well below the maximum acceptable daily intake level of 40-50 mg/kg body weight/day, careful consideration is warranted when consuming food products that may affect neurobehavioral health. © 2014 Wiley Periodicals, Inc.